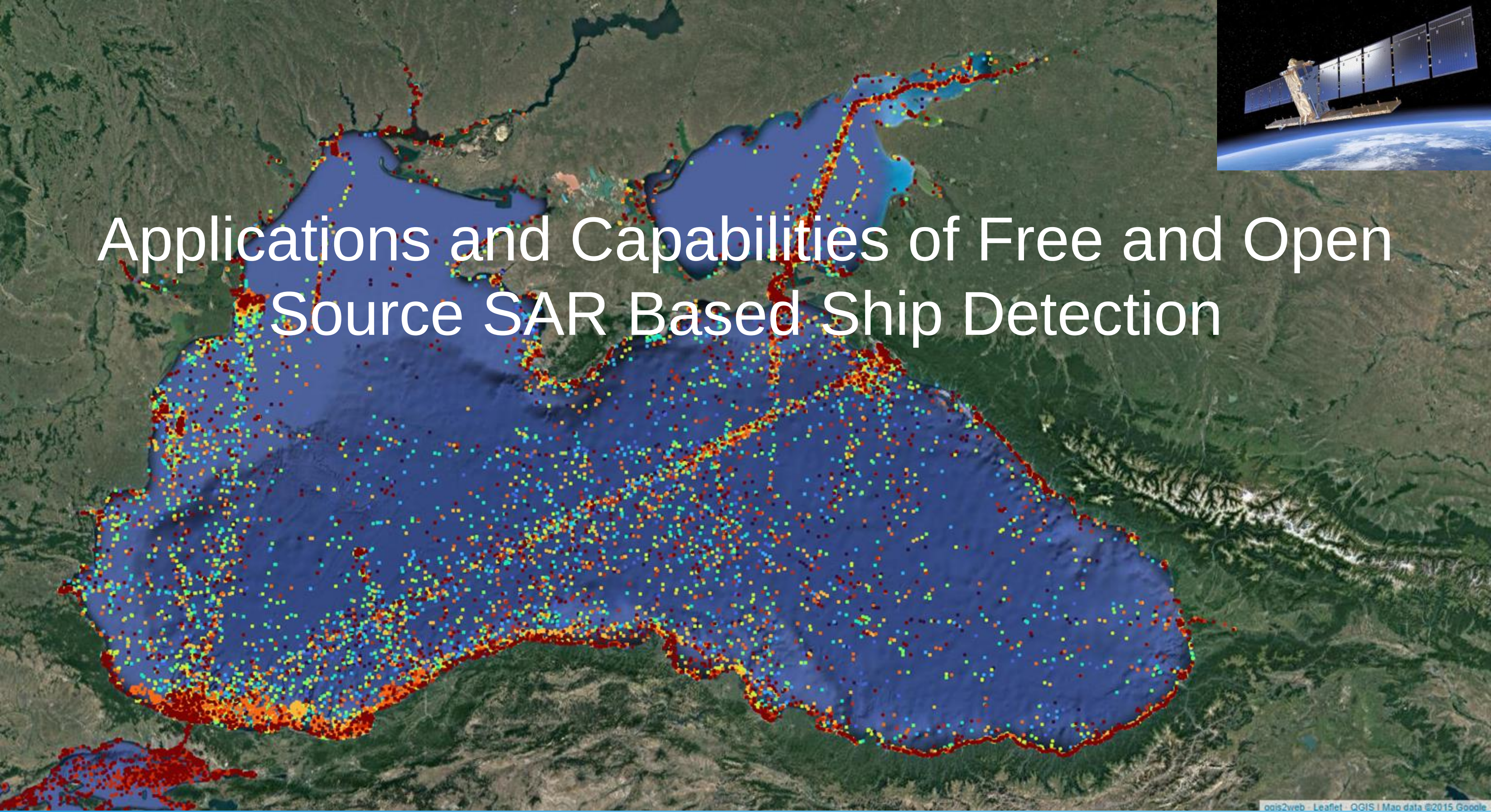
Applications and Capabilities of Free and Open Source SAR Based Ship Detection



Overview

- Motivation
- Methodology
- Results
- Next Steps

Motivation

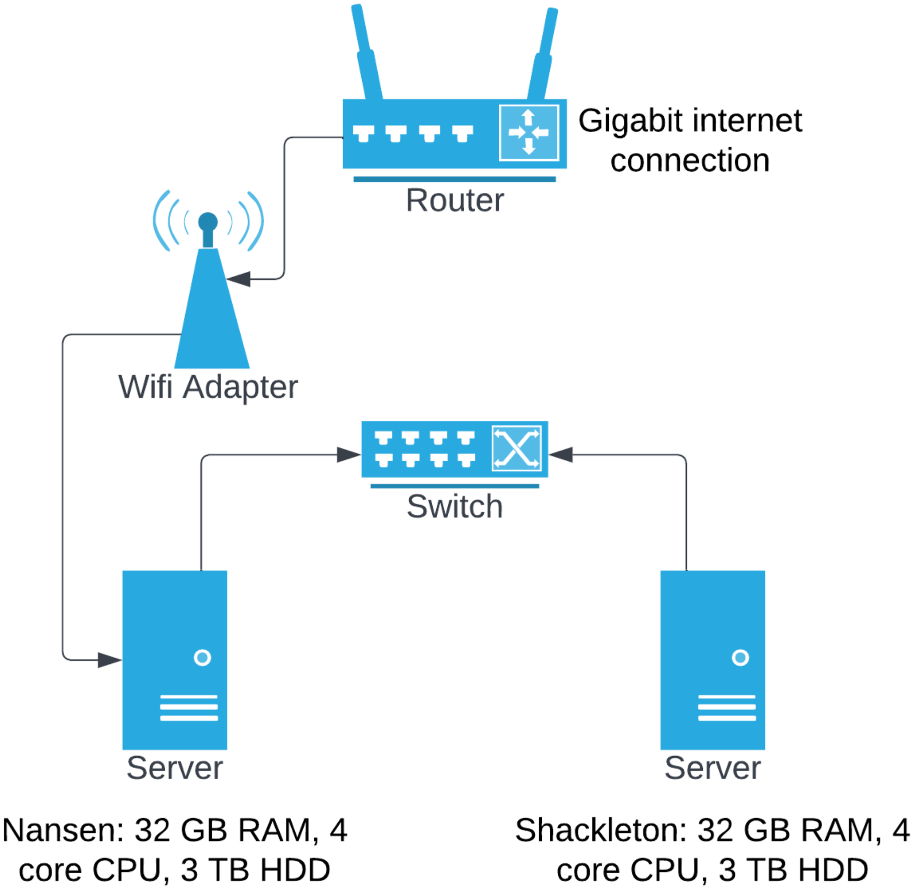
- All current ship tracking information is based on voluntarily broadcast AIS signals
- Issues with locating ships without transmitting transponders
- Illegal fishing
- Military operations
- Piracy
- Human trafficking

Current Applications

- Most applications in the motivation would require the integration and correlation of AIS data which is not currently implemented. Expected applications in its current state include:
  - Monitoring traffic in shipping lanes, port backlogs and other bottlenecks
  - Observing fisheries
- Unanticipated applications include:
  - Monitoring offshore wind farm construction
  - Offshore oil operations

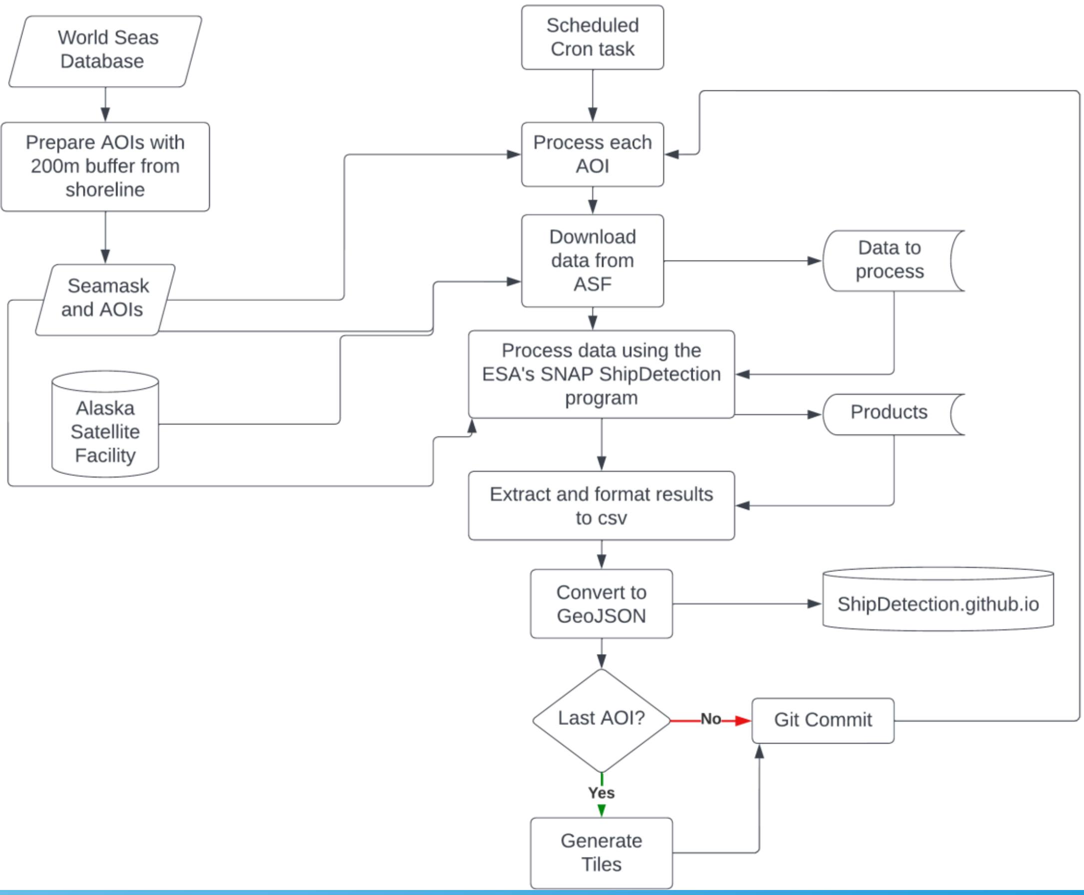
Processing Resources

- All processing occurs using 2 Lenovo Thinkcentres
- Processing AOIs shared between them and results output over the network to a single github repository to reduce risk of conflicts.
- Each with:
  - 32 GB RAM
  - 3TB HDD Storage
  - 4 core I5 CPU
  - Gigabit internet over a shared wifi adapter through a switch
- Currently takes ~8hrs a day for the processing to complete. Highly dependent on internet speed.



Processing Workflow

- Download data from the ASF
- Run Shipdetection algorithm using ESA snap with a landmask with a 200 m buffer
- Extract the results from output directory
- Convert results into leaflet compatible json
- Generate vector tiles
- Commit to webpage



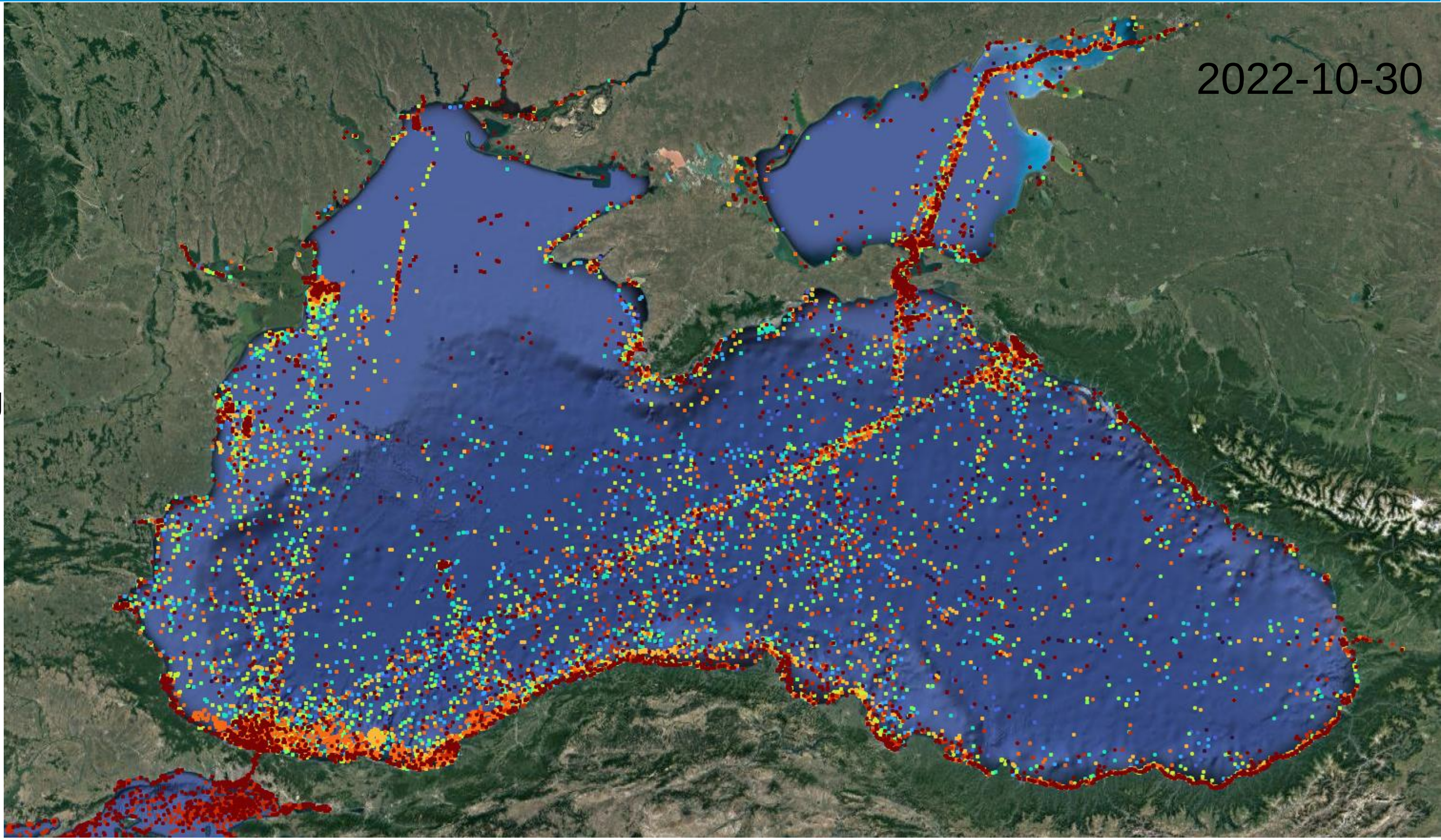
Results – Black Sea

- Colour scale based on age of the detection (Red to blue with 12 day increments up to 120 days. Based on up to 10 revisits of SNT

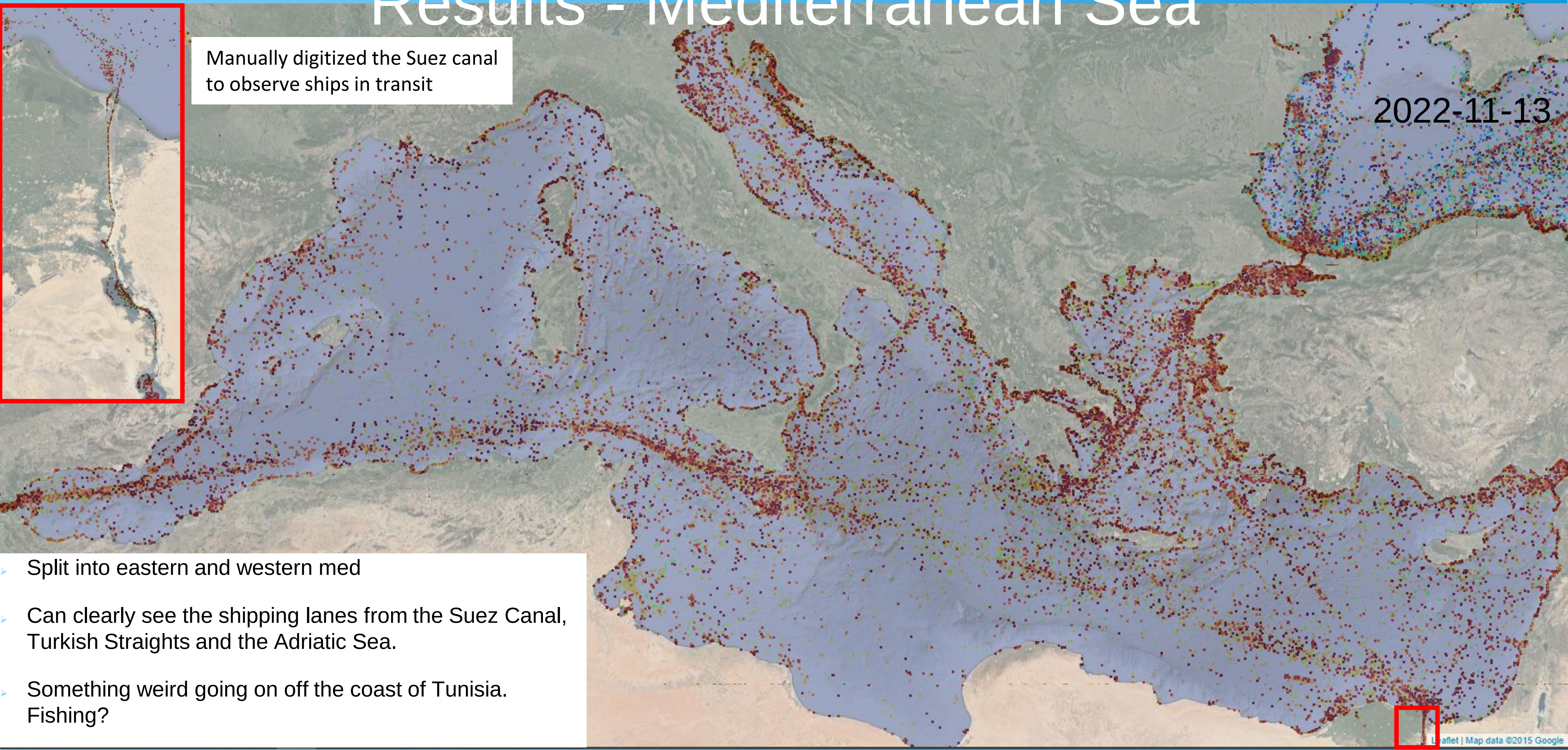
Notable observations:

- Reduction in shipping backlog around the Danube Delta originally caused due to the blockade of Odesa

- Reopening of shipping lanes to the port of Odesa following the grain deal



Results - Mediterranean Sea



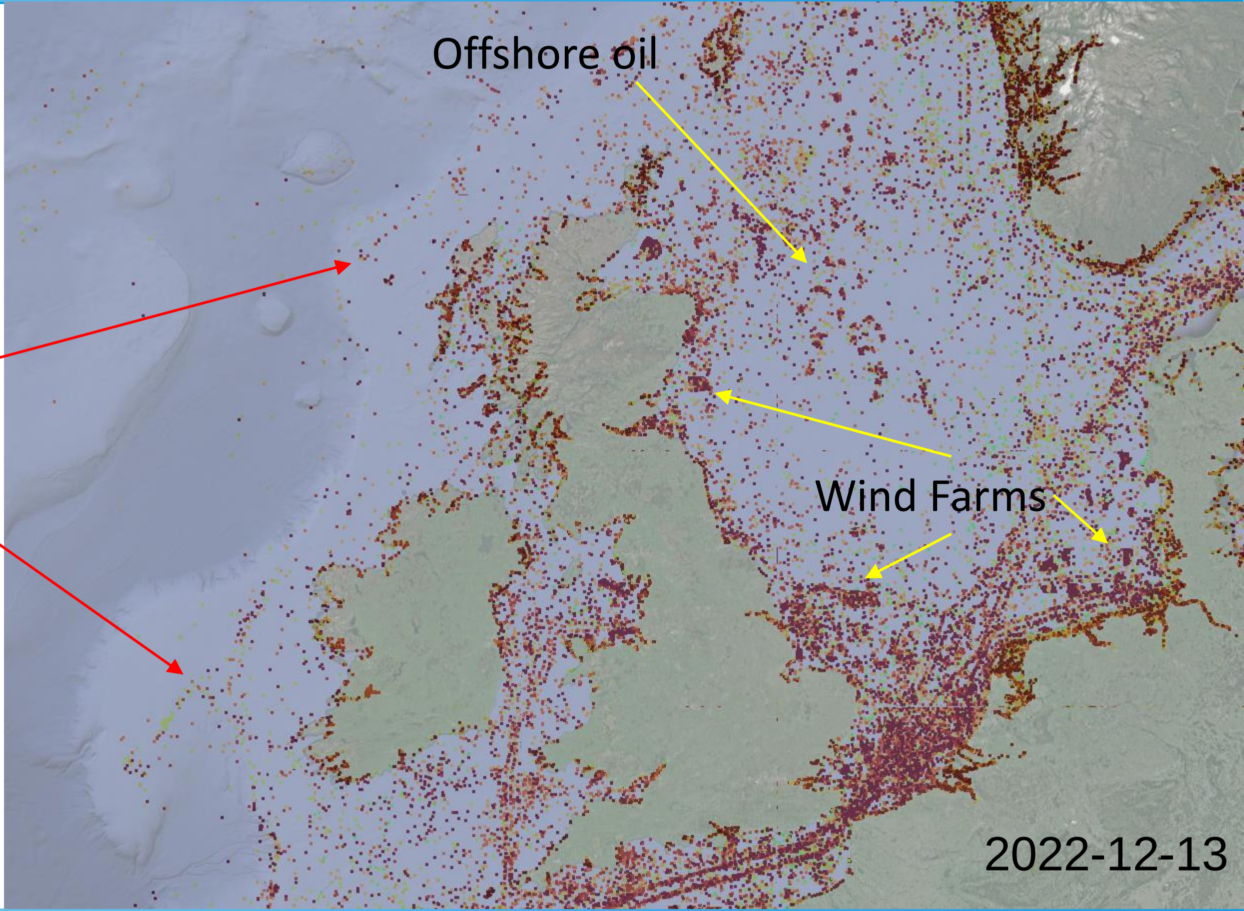
Manually digitized the Suez canal to observe ships in transit

- Split into eastern and western med
- Can clearly see the shipping lanes from the Suez Canal, Turkish Straights and the Adriatic Sea.
- Something weird going on off the coast of Tunisia. Fishing?

Results – North Sea

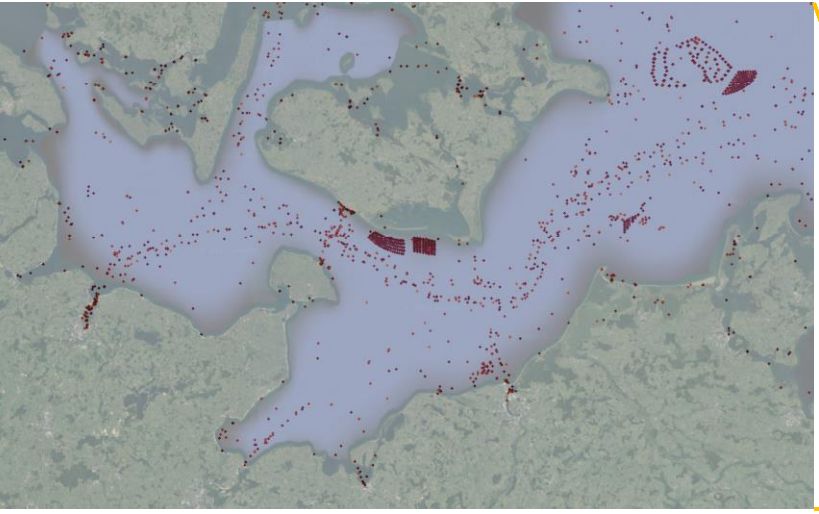
- Wind farms show up very well
- North Sea oil production highlighted
- Probable Location of fisheries indicated

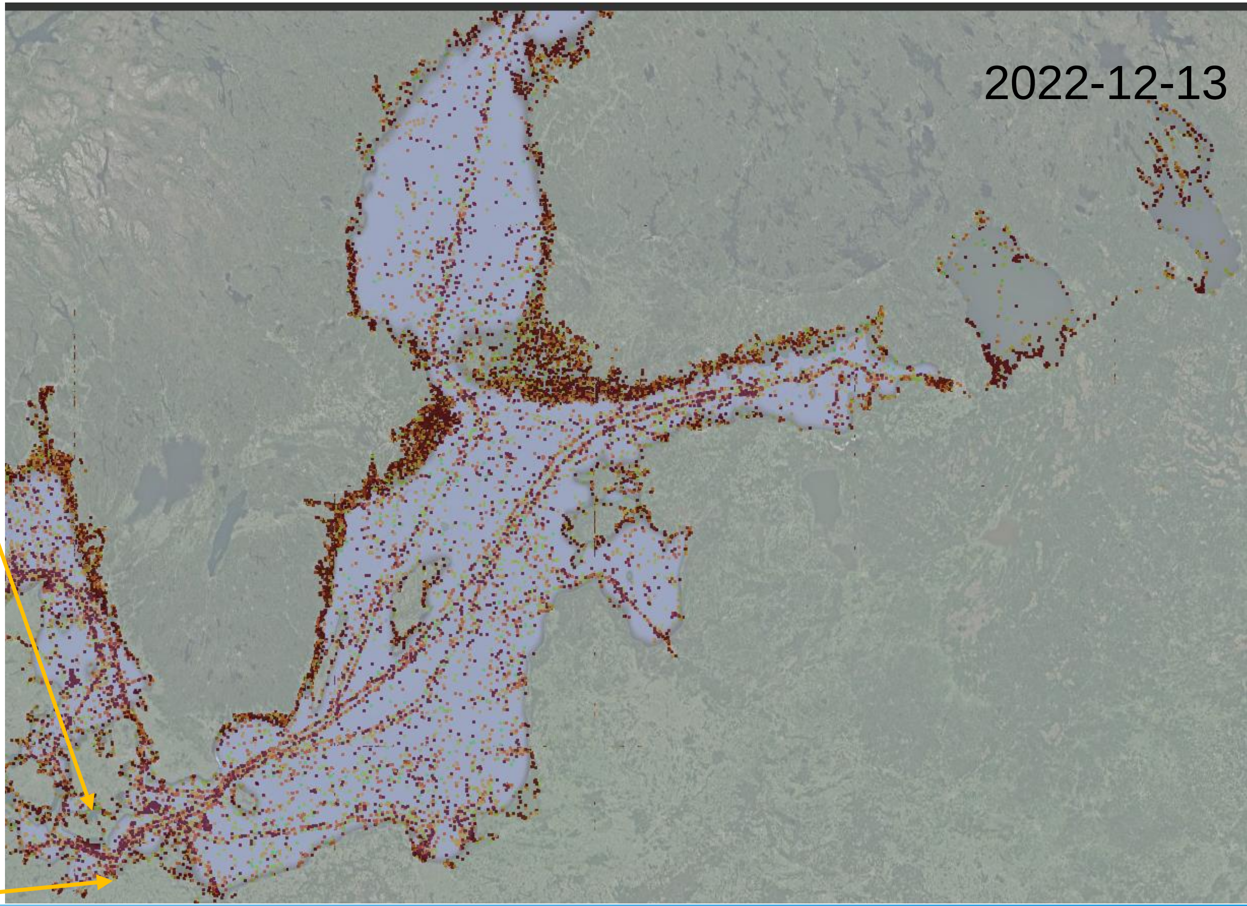
Probable
Fisheries



Results – Baltic Sea

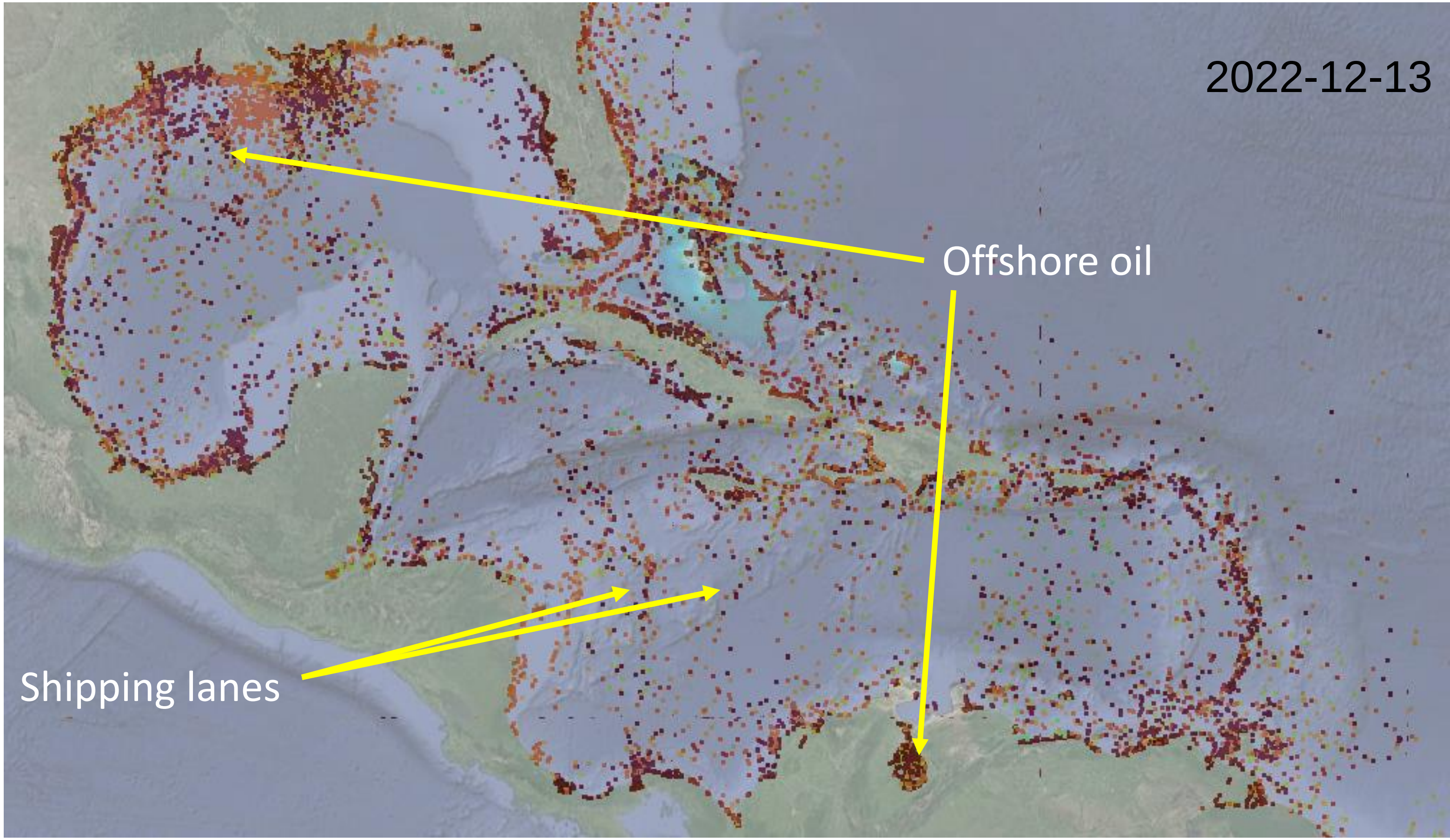
- Shows main shipping lanes
- Ferry routes and wind farms are apparent
- Lot of false positives off the coast of Sweden and Finland due to small rocky islands not included in the seamask





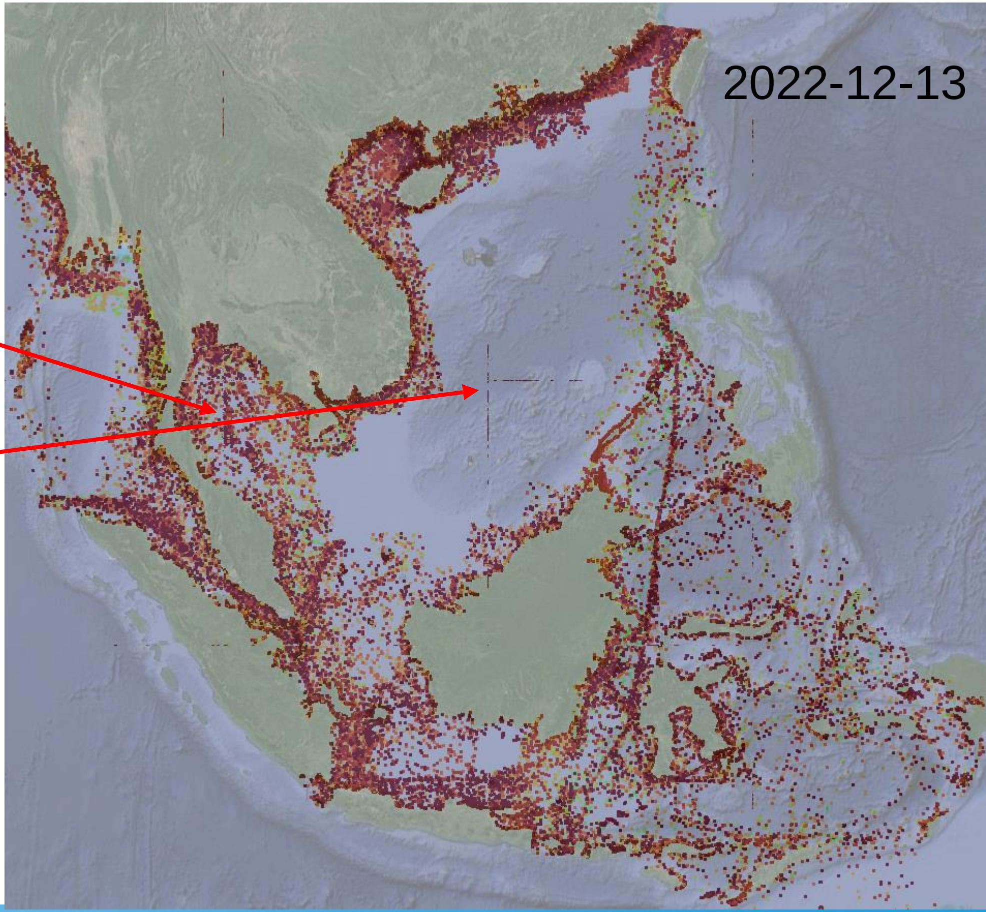
Results - Caribbean

- Offshore oil operations highlighted
- Shipping lanes to the Panama canal



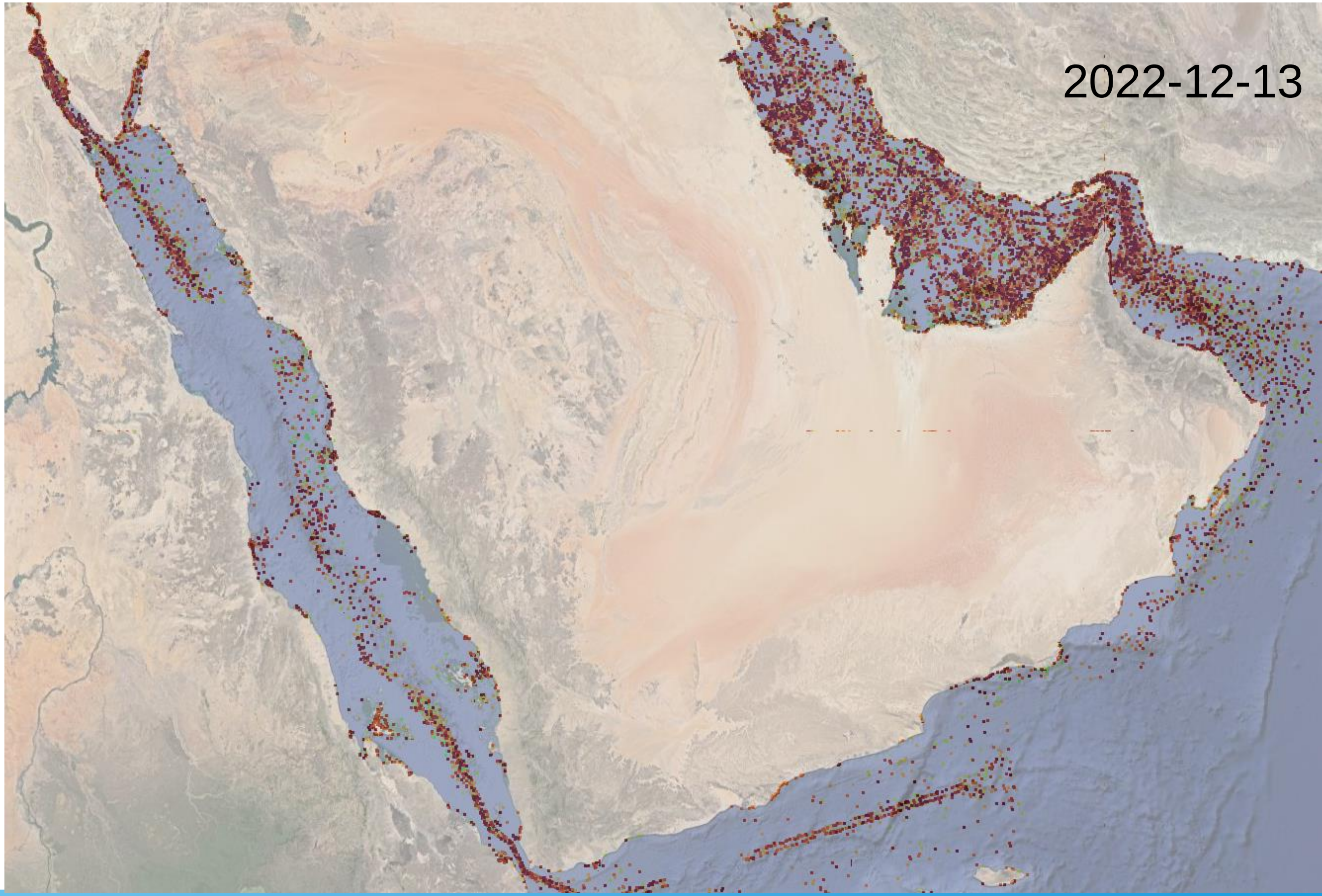
Results – Southeast Asia

- Main Shipping lanes are highlighted.
- Offshore oil in the Gulf of Thailand
- SNT is not acquired over a most of the South China Sea
- An issue is apparent here with edge effects on the borders of the tiles



Results – Arabian Peninsula

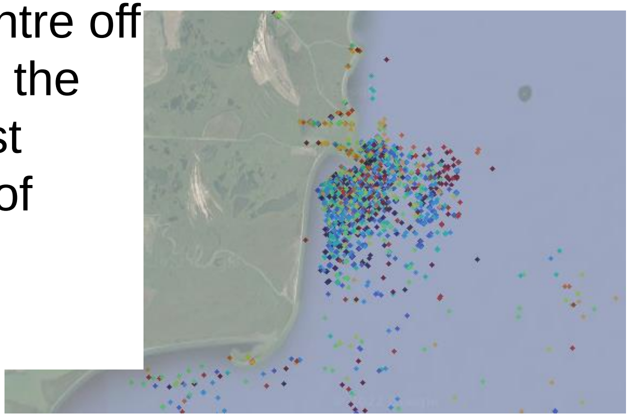
- SNT does not acquire over a portion of the Red Sea
- Shipping lanes from the Suez canal highlighted.
- Offshore oil in the Persian Gulf highlighted

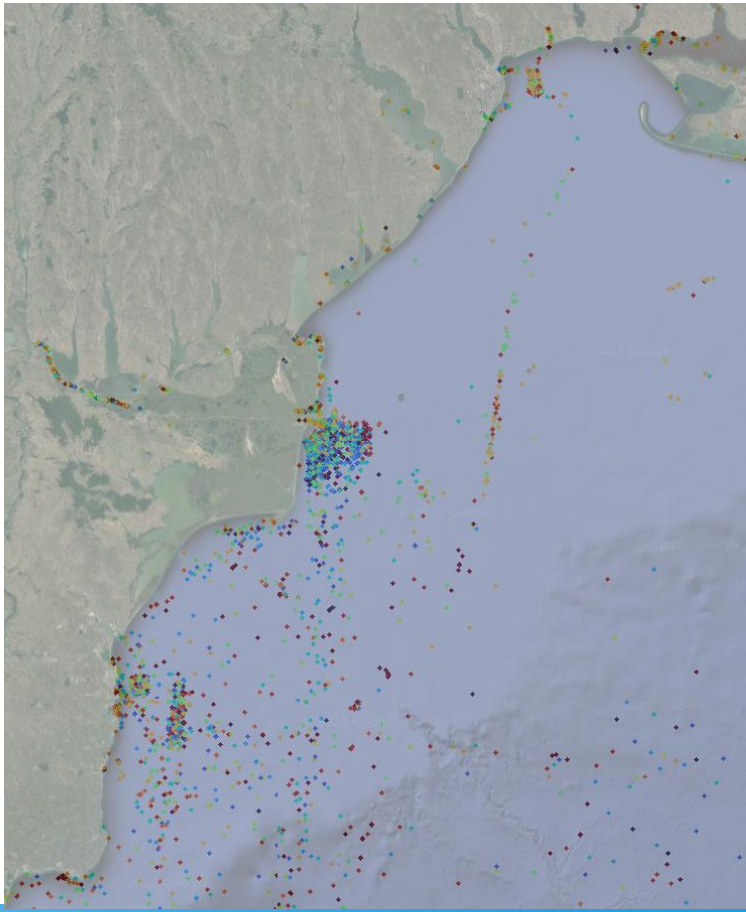


Applications – Changes to shipping routes

- Highlights changes to shipping activity
  - Good at indicating sudden backlogs, clearing of backlogs, increases and decreases in shipping lane usage

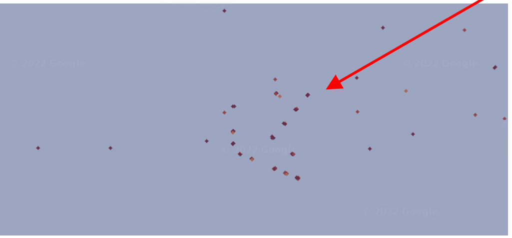
- Large area showing blue ships with a cluster of red ships in the centre off the coast of Romania, indicate the clearing of a backlog which first began with Russian blockage of Odesa





Applications – Offshore wind power

- Wind farms appear as regular grids of objects detected that never move
- Monitoring wind farm construction
- Currently under construction
- Can see construction ships at the locations indicated in the notice to mariners issued



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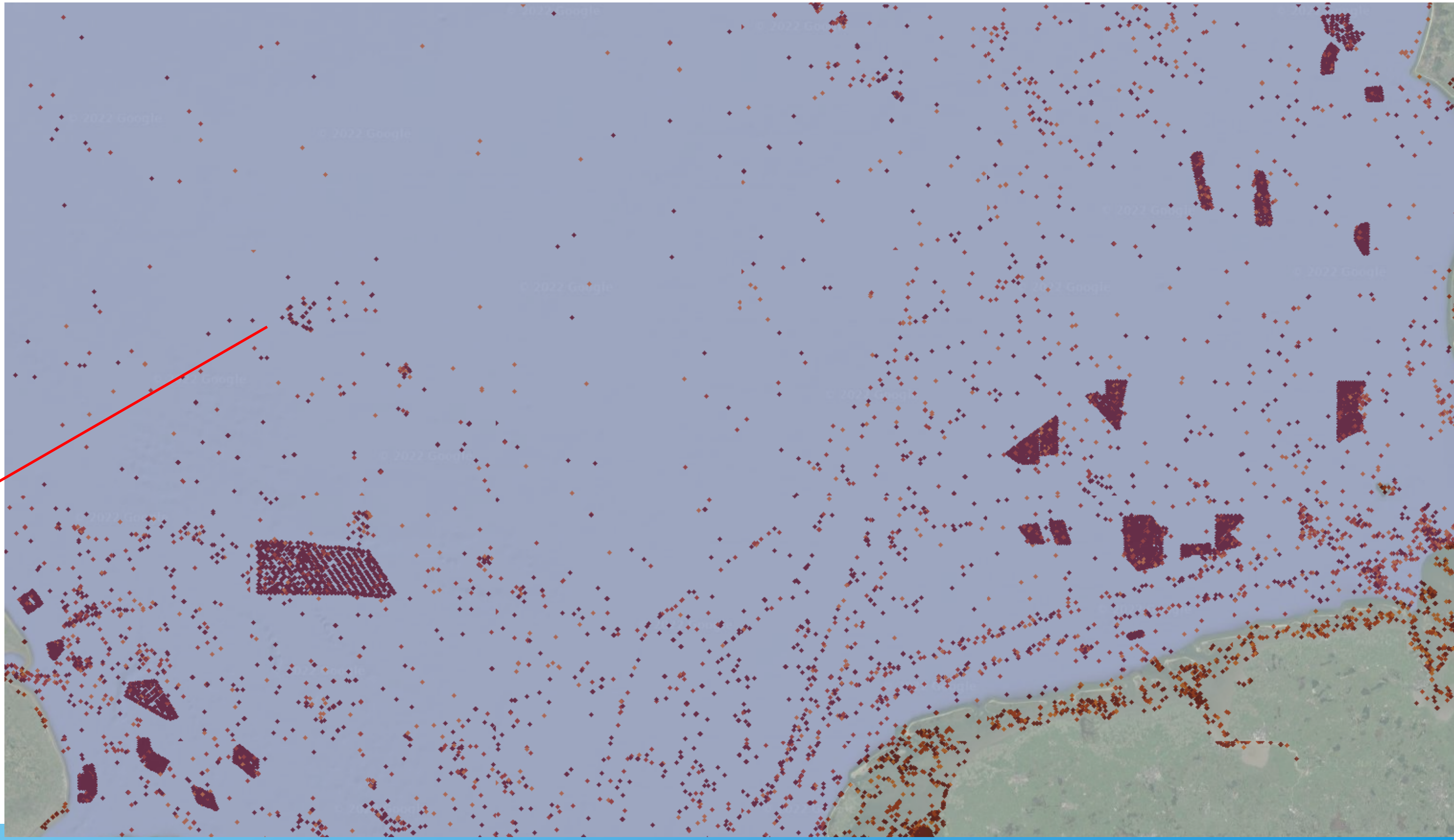
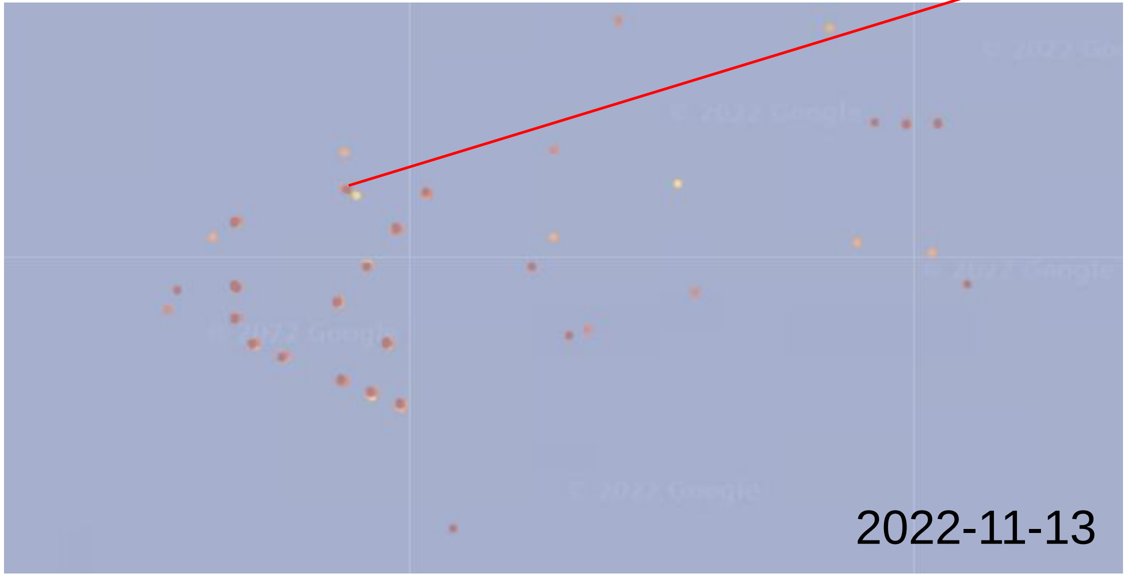
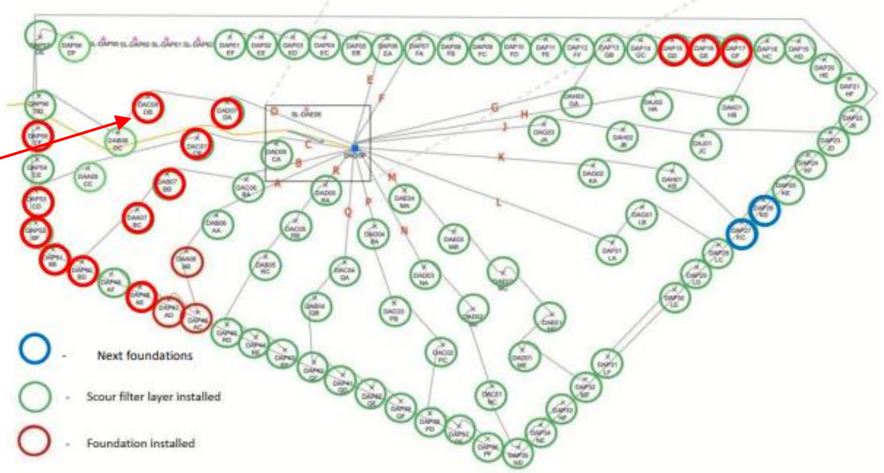


Applications – Offshore wind power

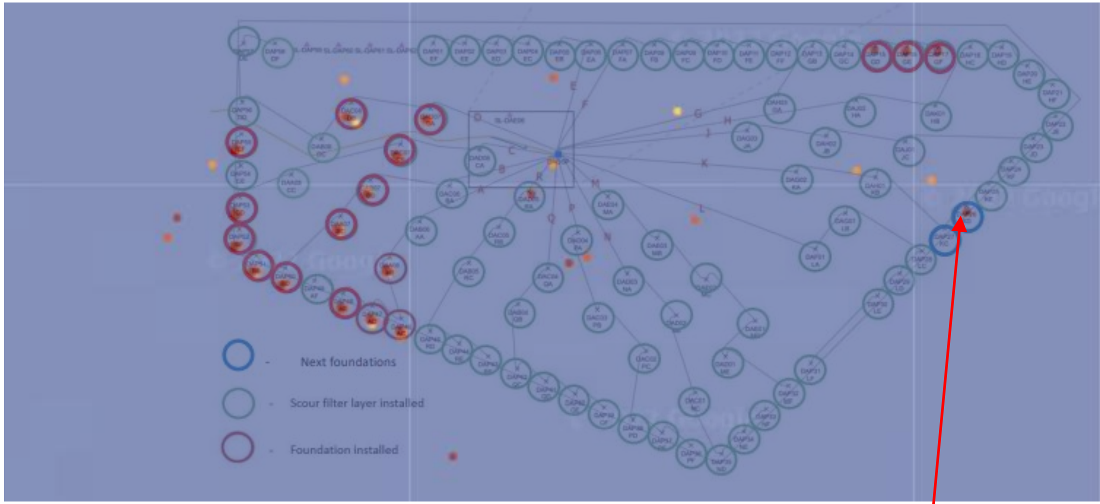
- Monitoring wind farm construction





Applications – Offshore wind power

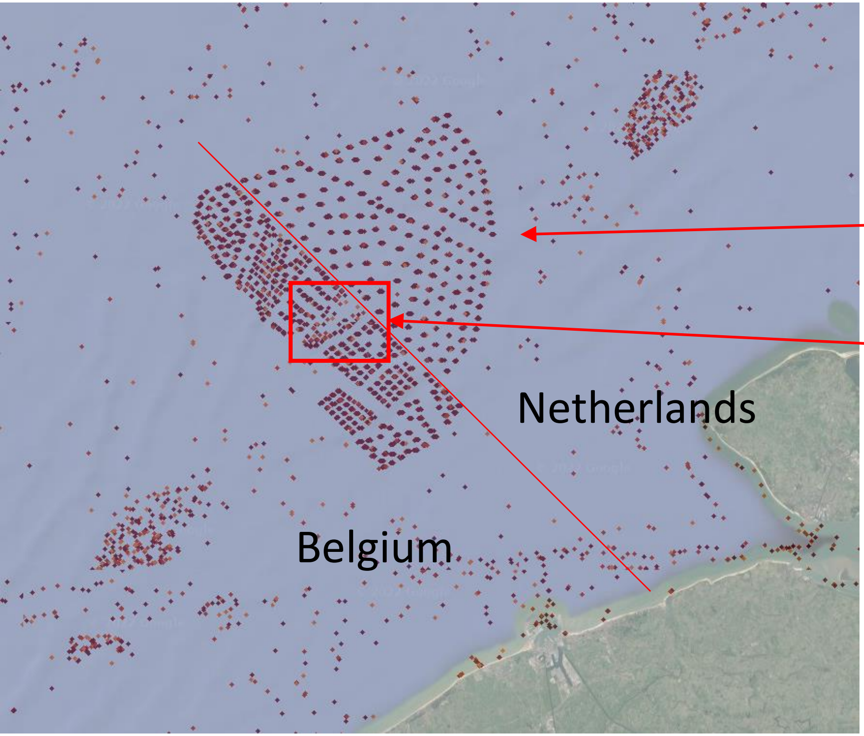
- Monitoring wind farm construction

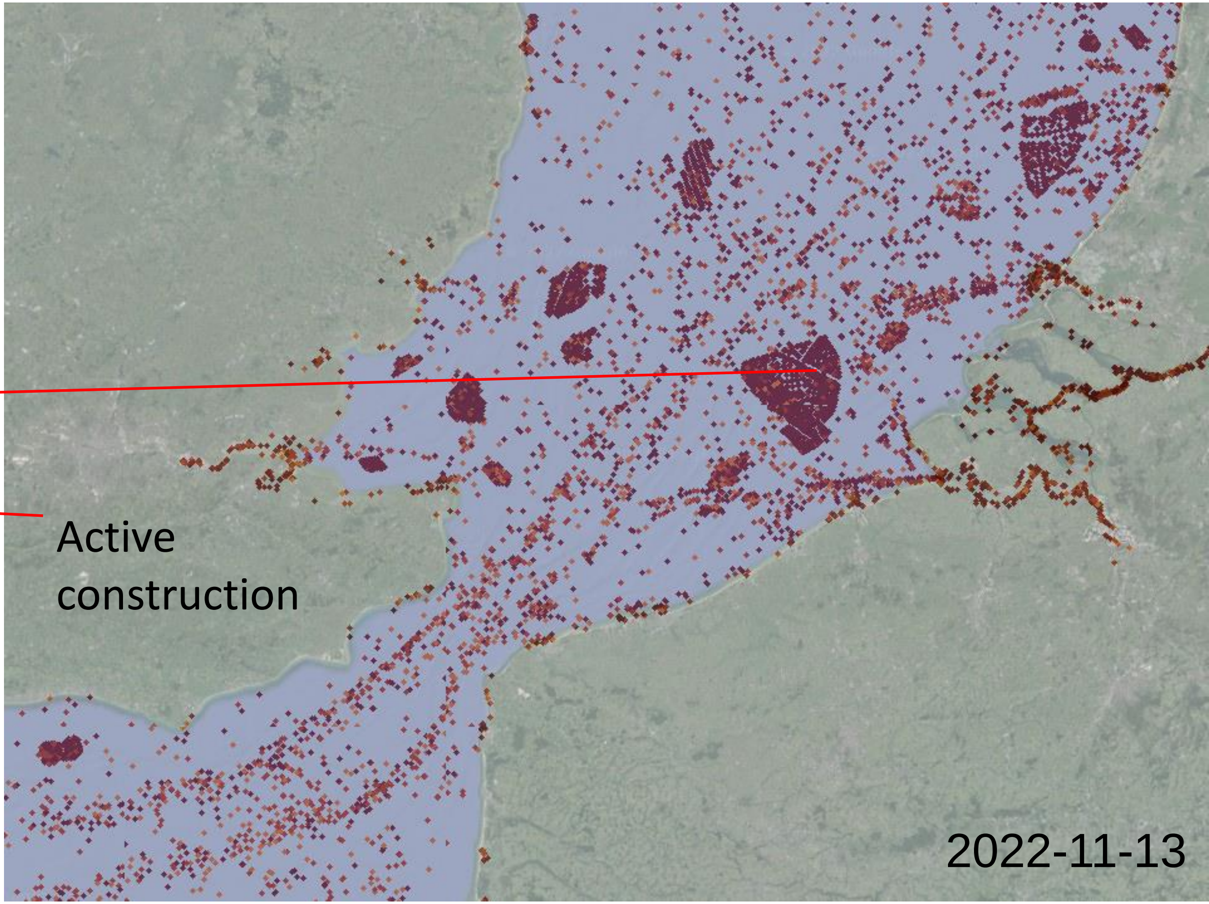


Construction seems to have progressed from “Next” to “In Progress”

Applications – Offshore wind power

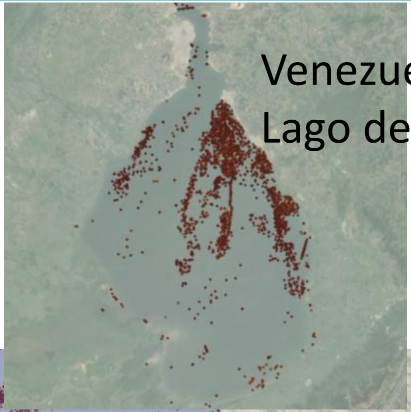
- Design differences between Belgian and Dutch wind projects

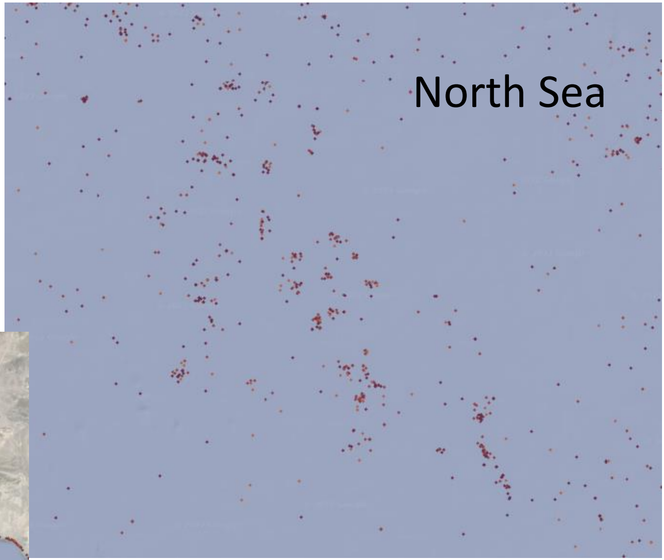


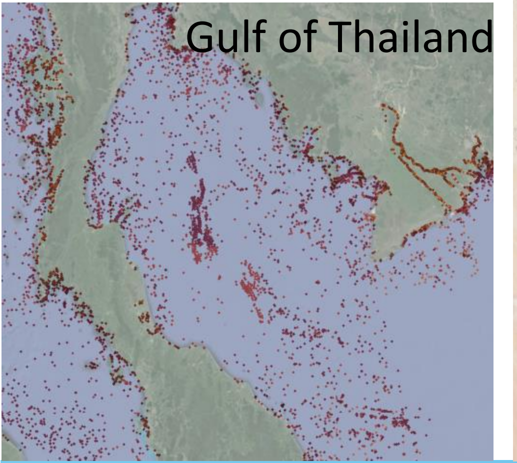


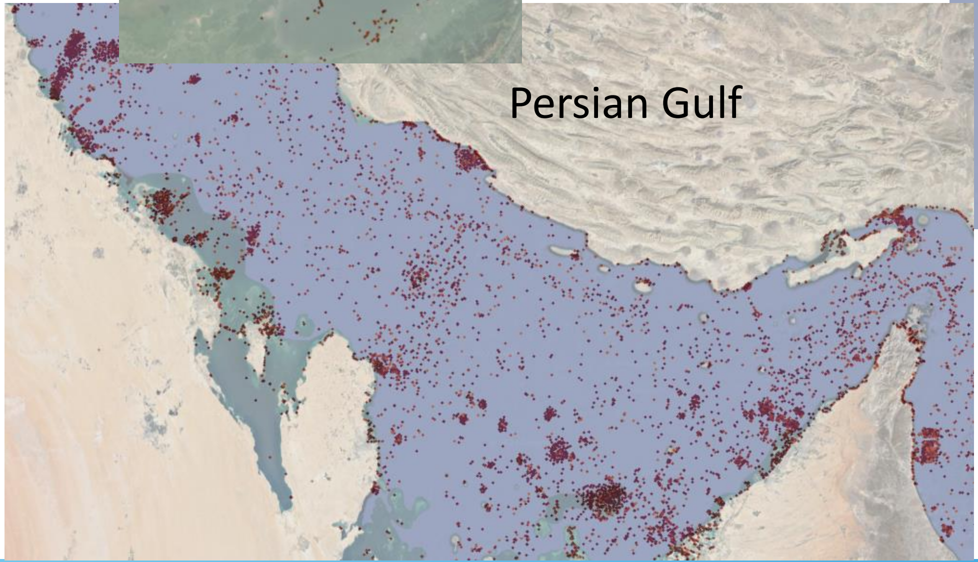
Applications – Offshore Oil Production

- Appear as irregular clusters of points around a static object
- Ships come and go around a static platform









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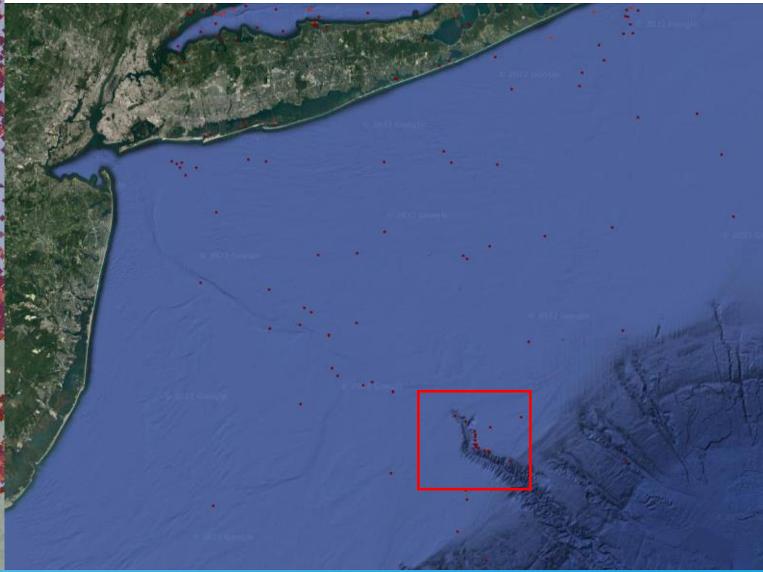
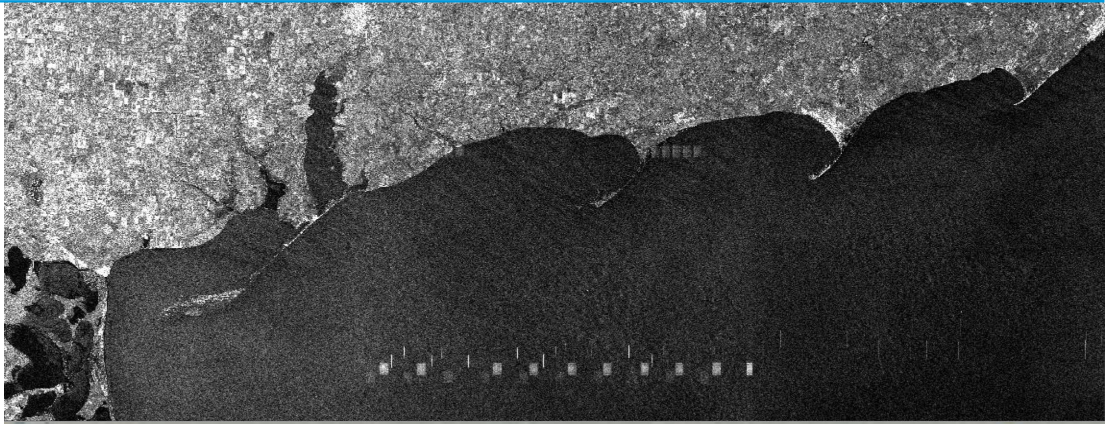
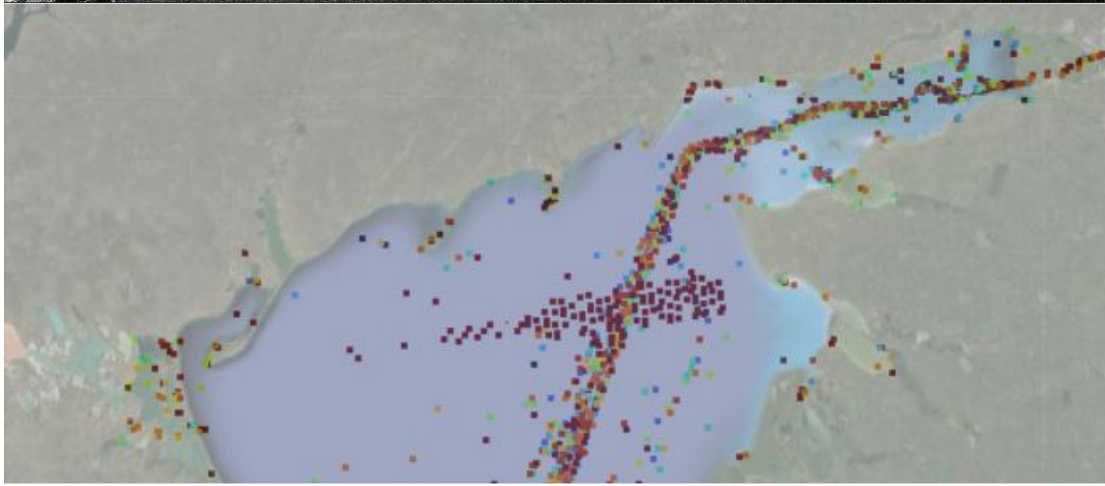
Applications – Fisheries

- Appear as clusters of ships, typically around a bank, always move between images



Applications – Active Radar Systems





- Active radar systems are known to cause interference in SAR imagery and since 2018 have been correlated with Air Defense systems according to a Bellingcat report.

- Appears in results as a distinct pattern of false detections

Next Steps

- Create simplified Sea Mask shapefiles so the download request can send a low vertex poly to query rather than a bounding box.
  - Will increase download efficiency so I don't download imagery over the middle of a continent or download imagery twice in overlapping bounding boxes.
  - This will allow me to include currently excluded AOIs in the South Indian and Pacific Oceans
- Include information on where SNT imagery is acquired
  - The user will see where no ships are located vs where no imagery is acquired.
- Include inland Seas and Large lakes (ie. Great Lakes and the Caspian Sea)
- Integrate other data sources to improve coverage and sampling
  - ie RCM, if possible to download by API

Acknowledgements

- Barry Carter, a Perl dev amongst men. Created the tiling script which makes the website usable. Without his help, it would have taken months more to get to where it is now.
- My friends and family who thought that this project was interesting enough to motivate me to not just give up when I ran into my first bug.
- My master’s supervisors Dr. Braun and Dr. Fotopoulos.
- My colleagues at TRE Altamira who also thought this project was cool and gave me a platform to share it.
- OpenAI and ChatGPT who cut months out of the learning process on how to build a website.